



Roundtable

Real-time multiplayer AI workspace platform.

Multiple users collaborate on AI conversations in real-time — with built-in tools for querying data warehouses, executing code, and managing files. Each workspace is an isolated container with its own AI, tools, and persistent storage.

Roundtable is designed as a **platform for agent orchestration** — connect your own A2A agents, MCP servers, or custom tools and let the AI route between them. Build agents in any language, deploy them anywhere, and plug them into a shared workspace where your whole team works together.

Quick Start

Zero config. No database. No `.env` file.

```
git clone https://github.com/foxtrotcommunications/foxtrotcommunication:
cd foxtrotcommunications-roundtable
npm install
npm run dev

# Open http://localhost:3000
```

Roundtable auto-detects the environment: if no `DATABASE_URL` is set, it uses SQLite and in-memory sessions for instant local development.

Three Deployment Paths

Path	Database	Command	Time	Use Case
Local dev	SQLite (auto)	<code>npm run dev</code>	~3 min	Development, testing, evaluation

Docker quickstart	PostgreSQL (compose)	<code>docker compose -f docker-compose.quickstart.yml up</code>	~3 min	Quick demo on a VM
GKE production	PostgreSQL (Cloud SQL)	<code>./deploy-gke.sh dev Development</code>	~5 min	Enterprise deployment

Features

- **Multiplayer AI Chat** — Multiple users in the same AI conversation, streaming in real-time
- **Concurrent Generation** — Each user has an independent AI generation lifecycle; multiple people can prompt AI simultaneously without blocking each other
- **Multi-Provider AI** — OpenAI, Anthropic (Claude), and Google Gemini (via Vertex AI or direct API key)
- **14 Built-in Tools** — Web search, data warehouse queries, file management, shell execution, and more
- **Configurable Tool Set** — Enable or disable individual tools per workspace via the Settings panel
- **Configurable Agent** — Set the AI provider, model, and system prompt per workspace — no redeploy needed
- **Data Warehouse Queries** — AI can query BigQuery, Snowflake, and Databricks in real-time
- **Workspace-per-Container** — Each workspace is an isolated container with its own identity
- **Workspace Bridges** — Open cross-workspace channels for AI-mediated collaboration between teams
- **A2A Agent Protocol** — Plug in external agents built in any language via the A2A standard
- **Multi-Cloud** — Deploy on Cloud Run, GKE, EKS, AKS, or any Kubernetes cluster
- **BYOK** — Bring Your Own Key; users configure their own API keys, or use server-level defaults

- **Presence** — See who's online in each workspace
- **Streaming** — AI responses stream token-by-token to all participants
- **Embeddable** — Embed in other apps via iframe with `EMBED_MODE=true`

Architecture

Browser (React + Socket.IO)

↕ WebSocket

Express + Socket.IO Server

↕

PostgreSQL

(sessions,
messages,
workspaces)

↕

AI Providers

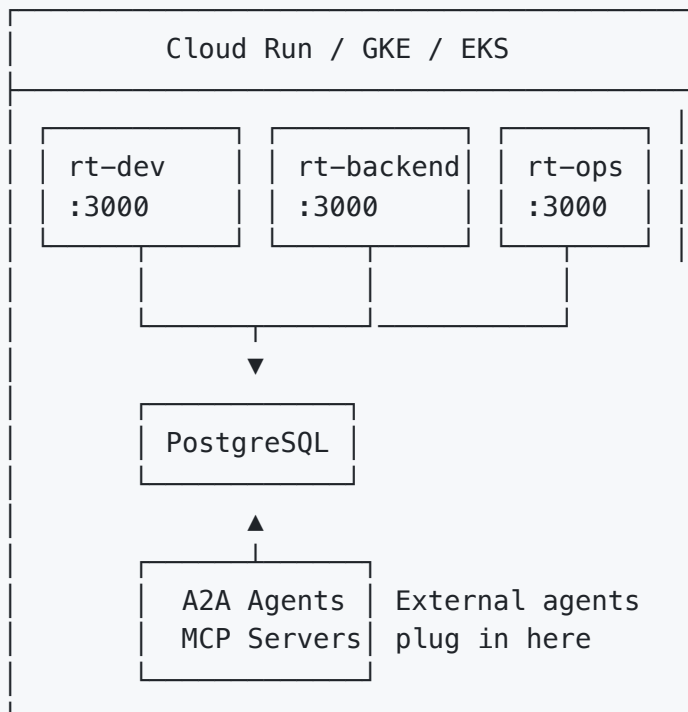
(OpenAI / Anthropic /
Vertex AI Gemini)

↕

Data Warehouses

(BigQuery / Snowflake /
Databricks)

Deployment model (workspace-per-container):



Each workspace = 1 container
All share the same PostgreSQL

Built-in Tools

All 14 tools are enabled by default. Individual tools can be toggled per workspace.

Tool	Description
<code>web_search</code>	Search the web via Google Custom Search or Vertex AI grounding
<code>read_url</code>	Fetch and extract text from web pages
<code>calculator</code>	Evaluate math expressions (powered by mathjs)
<code>run_code</code>	Execute JavaScript in a sandboxed environment
<code>query_bigquery</code>	Query Google BigQuery (read-only, max 100 rows)
<code>query_snowflake</code>	Query Snowflake (read-only, max 100 rows)
<code>query_databricks</code>	Query Databricks SQL Warehouse (read-only, max 100 rows)
<code>shell_exec</code>	Execute allowlisted shell commands in the workspace
<code>read_file</code>	Read files from the workspace directory
<code>write_file</code>	Write files to the workspace directory
<code>list_files</code>	List files in the workspace directory
<code>find_file</code>	Search for files by name
<code>git_clone</code>	Clone a git repository into the workspace
<code>git_commit</code>	Stage and commit changes

Data warehouse tools enforce **read-only access** — INSERT, UPDATE, DELETE, DROP, and other write operations are blocked at the tool level.

Configuration

Variable	Default	Description
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PORT	3000	Server port
SESSION_SECRET	dev default	Secret for session cookies (required in production)
DATABASE_URL	(none)	PostgreSQL connection string. If unset, uses SQLite for local dev
WORKSPACE_ID	default	Unique workspace identity
WORKSPACE_NAME	Roundtable	Display name for the workspace
EMBED_MODE	false	Allow iframe embedding

AI Providers

Variable	Description
OPENAI_API_KEY	Server-level OpenAI key
ANTHROPIC_API_KEY	Server-level Anthropic key
GOOGLE_AI_API_KEY	Server-level Google AI key
GCP_PROJECT	GCP project for Vertex AI (uses ADC, no key needed)
GCP_LOCATION	Vertex AI region (default: <code>us-central1</code>)

Data Warehouses

Variable	Description
GCP_PROJECT	BigQuery — uses same ADC as Vertex AI
SNOWFLAKE_ACCOUNT	Snowflake account identifier
SNOWFLAKE_USERNAME / PASSWORD	Snowflake credentials
DATABRICKS_HOST	Databricks workspace URL
DATABRICKS_TOKEN	Databricks personal access token

Workspace Settings

Each workspace can be configured at runtime via the **Settings panel** (no redeploy required):

- **AI Agent:** Provider, model, and system prompt
- **Tools:** Enable or disable individual tools per workspace
- **API Keys:** Users can configure personal API keys that override server defaults

Testing

```
npm test
```

98 tests across 8 suites covering shell execution security, SQL safety for all three data warehouse tools, file operations, authentication, and configuration parsing.

Deployment Options

Docker Quickstart (VM)

```
curl -O https://raw.githubusercontent.com/foxtrotcommunications/foxtrotc  
docker compose -f docker-compose.quickstart.yml up
```

GKE (Google Kubernetes Engine)

```
GCP_PROJECT=your-project ./deploy-gke.sh --setup  
GCP_PROJECT=your-project ./deploy-gke.sh dev Development
```

Any Kubernetes Cluster

```
kubectl apply -f k8s/base/config.yaml
kubectl create secret generic roundtable-secrets \
  --from-literal=SESSION_SECRET=$(openssl rand -hex 32) \
  --from-literal=DATABASE_URL="postgresql://user:pass@host:5432/roundtable"
./deploy-k8s.sh dev Development
```

License

Apache License 2.0

Built by [Foxtrot Communications](#)